KEY MODULE 1

SERENDIPITY IN ACTION!

SECTION 1: page 6

Dado que el estudiante debe interactuar con el texto de la Sección 1, muchas de las respuestas son de inferencia y/u opinión. Por lo tanto, la gran mayoría de respuestas que se presentan a continuación son a modo de ejemplo.

BEFORE READING

Serendipity /ˌsɛr(ə)nˈdɪpɪti/

c. the faculty or phenomenon of finding valuable or beneficial things by chance

WHILE READING

a. Discovery: anesthesia

   Discoverer: Horace Wells

b. Discovery: chewing gum

   Discoverer: Thomas Adams

c. Discovery: penicillin

   Discoverer: Alexander Fleming

AFTER READING

VOCABULARY BUILDING: page 8

a. 5
b. 3
c. 1
d. 2
e. 4

f. 7
g. 6
h. 8

GRAMMAR STOP: as, therefore, however

- In 1844, Horace Wells noted the pain-killing properties of nitrous oxide ("laughing gas") during a laughing-gas roadshow **as** a man gashed his leg but did not feel any pain.
- In 1844, Horace Wells noted the pain-killing properties of nitrous oxide ("laughing gas") during a laughing-gas roadshow **BECAUSE** a man gashed his leg but did not feel any pain.

- A man gashed his leg but did not feel any pain. **Therefore**, Wells started using nitrous oxide in painless dental operations.
- A man gashed his leg but did not feel any pain, **SO** Wells started using nitrous oxide in painless dental operations.

- Wells was exposed to ridicule as onlookers concluded that the patient felt pain. **However,** he did not give up on his hypothesis and nitrous oxide became an early form of anesthesia.
- Wells was exposed to ridicule as onlookers concluded that the patient felt pain, **BUT** he did not give up on his hypothesis and nitrous oxide became an early form of anesthesia.

The CONNECTORS table						
Addition	**Result**	**Contrast**	**Reason**	**Time**	**Condition**	**Position and punctuation**
and	so	but				**(1)** __ ~ __ . or __ , ~ __ .
	THEREFORE	HOWEVER				**(2)** __ . ~ , __ . or __ ; ~ , __ .
			because AS	when	if	**(3)** ~ __ , __ . or __ ~ __ .

CHECK YOUR PROGRESS 1: page 11

a)

1. As
2. therefore
3. However
4. as

b)

1. As
2. Therefore
3. however
4. as

c)

1. Penicillin destroys the bacteria's cell walls. Therefore, it is used for bacterial infections.

   Penicillin destroys the bacteria's cell walls; therefore, it is used for bacterial infections.

   Penicillin is used for bacterial infections as it destroys the bacteria's cell walls.

   As penicillin destroys the bacteria's cell walls, it is used for bacterial infections.

2. Human cells do not have cell walls. Therefore, penicillin does not harm them.

   Human cells do not have cell walls; therefore, penicillin does not harm them.

   Penicillin does not harm human cells as they do not have cell walls.

   As human cells do not have cell walls, penicillin does not harm them.

3. Penicillin G is widely used. However, it is not resistant to digestive acids, which limits its usefulness.

   Penicillin G is widely used; however, it is not resistant to digestive acids, which limits its usefulness.

SECTION 2

CHECK YOUR PROGRESS 2: page 13

a)

1. are still used
2. passess
3. must be given
4. is usually given
5. resists
6. can be given

b)

1. is used
2. was foreseen
3. was contracted
4. was left
5. was patented

c)

1. is recognized
2. observed
3. could be used
4. was given
5. assumed

EXPRESS YOURSELF: page 15

b)

Discovery, inventor and year	What happened	Big discovery	Result
Saccharin Constantin Fahlberg and Ira Remsen, 1879	Fahlberg spent the day studying coal tar derivatives at Johns Hopkins laboratory. He left the place and went to dinner.	Something he ate tasted particularly sweet. He discovered a chemical on his hand that tasted like sugar. It turned out to be calorie-free.	He patented the breakthrough, saccharin. He did so secretly. He cut Remsen and the university out of millions of dollars.
Botox Alastair and Jean Carruthers, 1987	The Carruthers were using small doses of a deadly toxin to treat eye-muscle disorders.	The medicine could reduce wrinkles. They were curious. They decided to continue studying this effect.	They learned it was safe. The medicine became Botox. Many

	They noticed an interesting side effect.		people use it to look younger.

Paragraphs: Students' own productions.

c)

En esta actividad se deberá desplegar e integrar todo el conocimiento de la lengua adquirido hasta el momento. Para su corrección, el docente le indicará el modo de presentación.

SECTION 3

QUESTION BLOCK: page 19

a and b)

In 1870, Thomas Adams tried to use chicle, a sap from a Mexican tree, to make rubber. His idea did not work. However, he found that chicle could be used to make chewing gum. People liked this new product very much. Therefore, (2) Adams started (1) the American Chicle Company in 1899. Today, (4) the chewing gum industry makes (3) billions of pieces of gum every year.

1. What did Adams start in 1899?
2. Who started the American Chicle Company?
3. What does the chewing gum industry make today?
4. Who makes billions of pieces of gum every year?

OBJECT QUESTIONS

INTERROGATIVE WORD	AUXILIARY (DO-DOES-DID)	ACTIVE SUBJECT	MAIN VERB Infinitive	…?

SUBJECT QUESTIONS

INTERROGATIVE WORD (SUBJECT)	CONJUGATED VERB	…?

CARDS A AND B

Today, [A]doctors use anesthesia to help patients feel no pain, but this idea started [B]in the mid-1800s. In [1]1844, dentist Horace Wells saw a man who hurt himself but did not feel pain after he inhaled nitrous oxide. Wells thought [2]that the gas could help his patients, so he tested the gas on [C]himself. The experiment worked, and [3]Wells believed that this substance could prevent pain. [D]His discovery made dental treatment easier and safer. Today, [4]historians consider him one of the first people to use anesthesia in dentistry.

1. When did Horace Well see a man who hurt himself but did not feel pain?
2. What did Wells think?
3. Who believed that this substance could prevent pain?
4. Who considers him one of the first people to use anesthesia in dentistry.

A. Who uses anesthesia to help patients feel no pain?
B. When did this idea start?
C. Who did he test the gas on?
D. What made dental treatment easier and safer?

QUESTIONS IN THE PASSIVE VOICE

a)

1. Who was allowed to demonstrate the use of nitrous oxide in 1845?
2. When was he allowed to demonstrate the use of nitrous oxide?

(1) <u>Horace Wells</u> was allowed to demonstrate the use of nitrous oxide (2) <u>in 1845</u>.

PASSIVE VOICE: OBJECT QUESTIONS

INTERROGATIVE WORD	AUXILIARY (Conjugated form of "be")	PASSIVE SUBJECT	MAIN VERB Past Participle	...?

PASSIVE VOICE: SUBJECT QUESTIONS

INTERROGATIVE WORD (SUBJECT)	AUXILIARY (Conjugated form of "be")	MAIN VERB Past Participle	…?

b)

1. a. What was covered with bacteria?
   b. What was the dish covered with?

When Alexander Fleming returned from his vacation, he found that (a) the dish was covered with (b) bacteria except in the area where there was mold.

2. a. What was chicle used for?
   b. What was used to manufacture chewing gum?

(b) Chicle was used (a) to manufacture chewing gum.

3. a. Who was exposed to ridicule when the patient groaned?
   b. When was Wells exposed to ridicule?
   c. What was Wells exposed to when the patient groaned?

(b) When the patient groaned, (a) Wells was exposed (c) to ridicule.

CHECK YOUR PROGRESS 3: page 21

a)

[a]Anesthesia helps doctors perform surgeries without pain, and it is used [b]in hospitals around the world. Today, patients usually receive local or general anesthesia [c]before operations. In the past, doctors used [d]ether in the first medical procedures, and the first public demonstration was performed [e]in 1846. .

.

a. What helps doctors perform surgeries without pain?
b. Where is anesthesia used?
c. When do patients usually receive local or general anesthesia?
d. What did doctors use in the past?
e. When was the first public demonstration performed?

2)

Chewing gum contains sweeteners and flavors, and people use it for enjoyment or fresh breath. Today, many companies produce [a]sugar-free gum, and it is sold [b]in almost every country. In the past, ancient civilizations chewed [c]natural tree resin, and the first modern chewing gum was created [d]in the 19th century.

a. What do many companies produce?
b. Where is chewing gum sold?
c. What did ancient civilizations chew?
d. When was the first modern chewing gum created?

3)

a. What did Fleming do halfway through an important experiment?
b. Why did he leave a dirty Preti dish in the lab sink?
c. What did he find when he returned?

EXPRESS YOURSELF: page 23

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TIME FOR FICTION I: page 26

VOCABULARY BUILDING

1. check
2. alive
3. shoot
4. face
5. breathless
6. couldn't believe his eyes
7. fine
8. scare
9. silly
10. hunter
11. tear up
12. game

AFTER READING

a)

4. He looked enthusiastic about his journey.

2. He met his guide.

1. He paid for the service.

3. He talked about the Machine and the presidential election.

b)

1. The company offered a time safari.
2. Eckels was promised to see dinosaurs.
3. If he broke the rules, he would pay another ten thousand dollars and face possible government actions.
4. He was breathless and couldn't believe his eyes.
5. Eckels felt satisfied.
6. Because six safari guides and twelve hunters were killed last year.

EXPRESS YOURSELF: page 27

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